SOMAIYA
VIDYAVIHAR
K J Somaiya Institute of Technology
An Autonomous Institute Permanently Affiliated to the University of Mumbai

DEPARTMENT OF INFORMATION TECHNOLOGY

Course Name and Code: Data Structures Lab **(**ITL302)
Semester: III (SYIT)
Academic Year: 2026-27 (Odd Semester)
Name: Rashi Dawda
Roll no: 23

Experiment No. 03

Aim: To implement the conversion of an Infix expression into a Postfix expression using a Stack and apply it to a real-world Heatwave Monitoring System for processing temperature-related calculations.

Objectives:
1.To understand the concept of Infix and Postfix expressions.
2.To implement a Stack for expression conversion.
3.To convert an Infix expression into its corresponding Postfix form.
4.To understand the use of operator precedence and associativity.
5.To apply expression conversion to a real-world calculation scenario.

Theory: An Infix expression is an expression in which the operator is written between two operands.

Example:

A + B

A Postfix expression is an expression in which the operator is written after its operands.

Example:

A B +

Infix-to-Postfix conversion is commonly performed using a Stack. During conversion, operands are directly added to the postfix expression, while operators are temporarily stored in the stack according to their precedence.

Common operator precedence is:

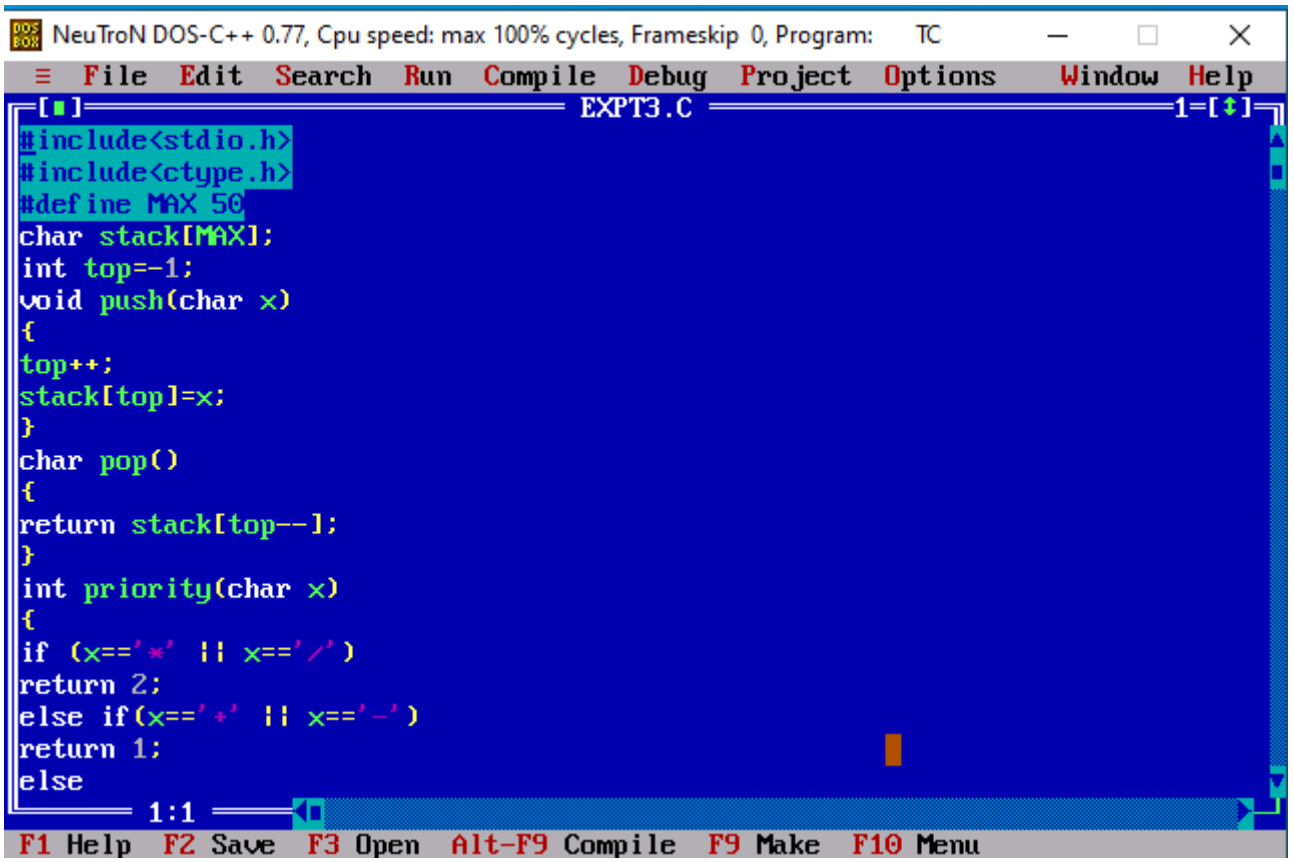
Operator	Precedence
^	Highest

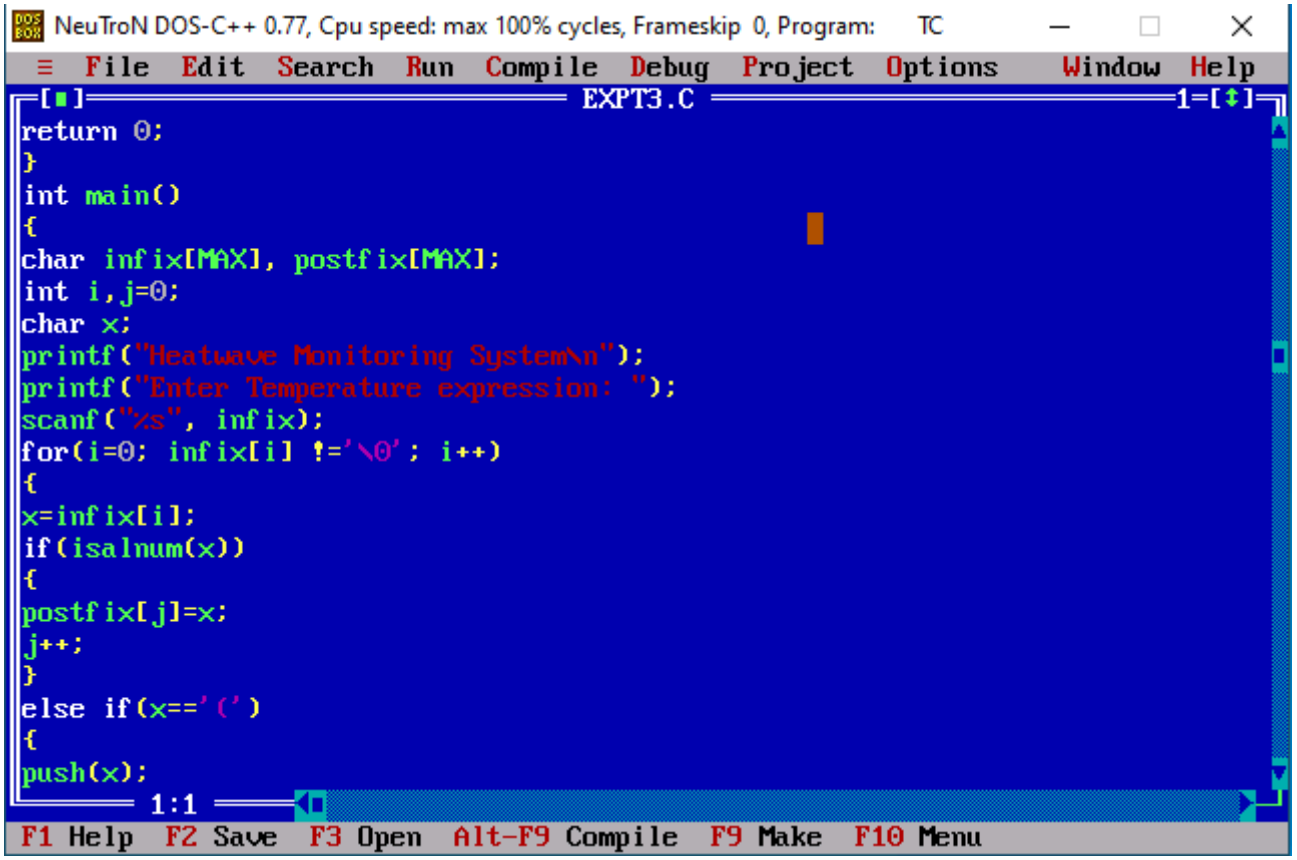
* / %	Medium
+ -	Lowest

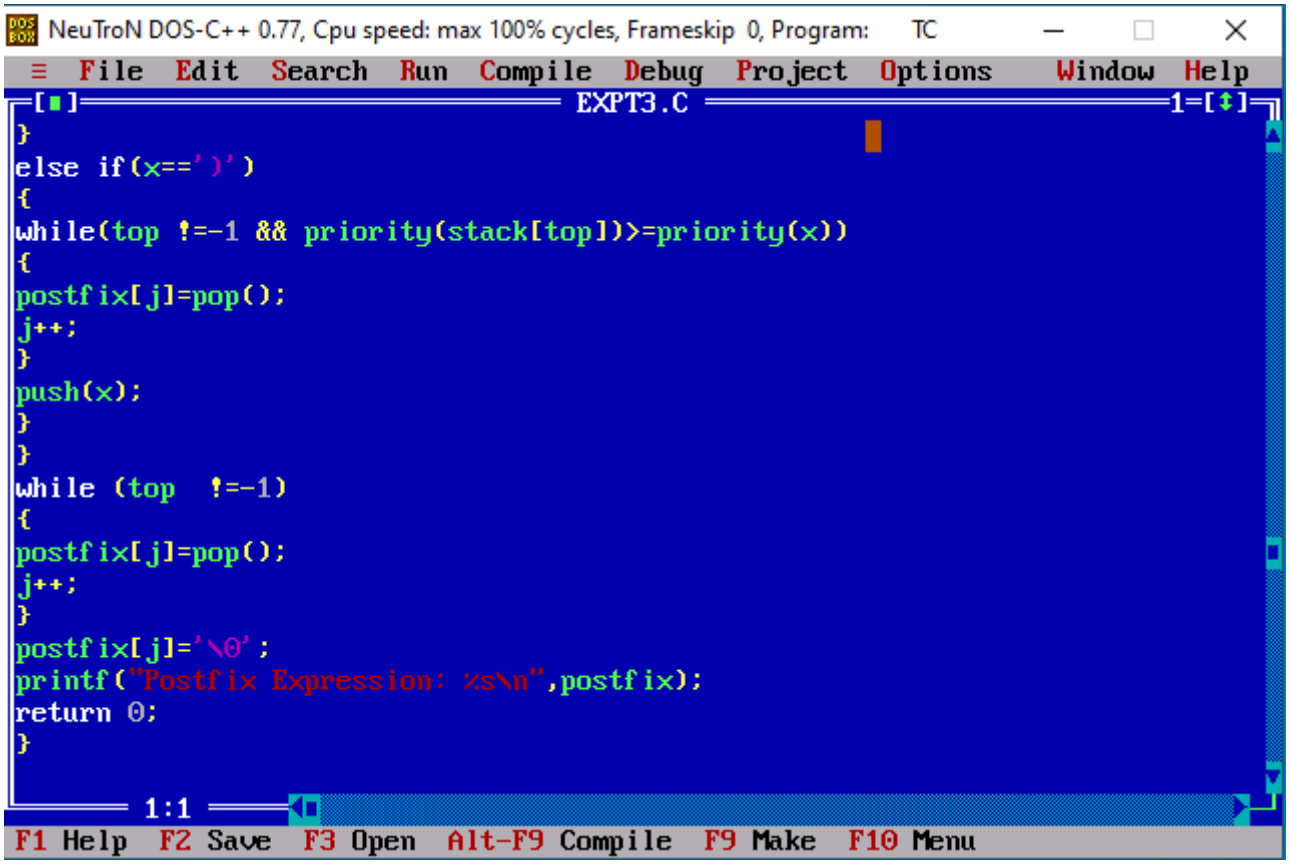
For example:

Infix: A + B * C
Postfix: A B C * +

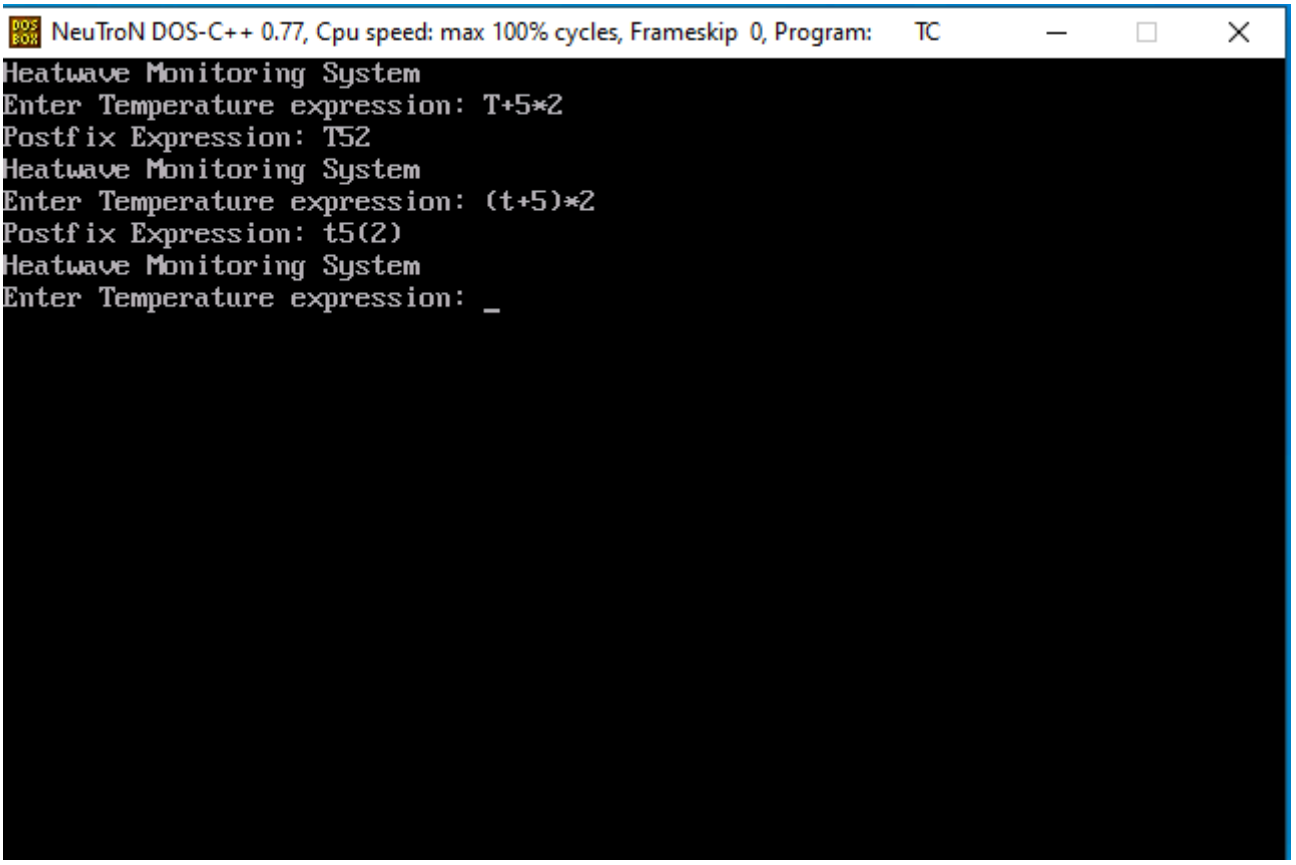
Code:




NeuTroN DOS-C++ 0.77, Cpu speed: max 100% cycles, Frameskip 0, Program: TC
File Edit Search Run Compile Debug Project Options Window Help
EXPT3.C
return 0;
}
int main()
{
char infix[MAX], postfix[MAX];
int i,j=0;
char x;
printf("Heatwave Monitoring System\n");
printf("Enter Temperature expression: ");
scanf("%s", infix);
for(i=0; infix[i] !='\0'; i++)
{
x=infix[i];
if(isalnum(x))
{
postfix[j]=x;
j++;
}
else if(x=='(')
{
push(x);
1:1
F1 Help F2 Save F3 Open Alt-F9 Compile F9 Make F10 Menu


NeuTroN DOS-C++ 0.77, Cpu speed: max 100% cycles, Frameskip 0, Program: TC
File Edit Search Run Compile Debug Project Options Window Help
EXPT3.C
}
else if(x==')')
{
while(top !=-1 && priority(stack[top])>=priority(x))
{
postfix[j]=pop();
j++;
}
push(x);
}
}
while (top !=-1)
{
postfix[j]=pop();
j++;
}
postfix[j]='\0';
printf("Postfix Expression: %s\n",postfix);
return 0;
}
1:1
F1 Help F2 Save F3 Open Alt-F9 Compile F9 Make F10 Menu

Output:



Conclusion: The Infix-to-Postfix conversion was successfully implemented using the Stack data structure. The experiment demonstrated how operators and operands are processed according to their precedence and how a complex expression can be converted into a form that is easier for a computer to evaluate. The concept can be applied to real-world systems such as heatwave monitoring calculations, where temperature and environmental data may require mathematical processing.

Outcome: After completing this experiment, the student will be able to:

- Understand Infix and Postfix expressions.
- Implement a Stack-based expression conversion.
- Apply operator precedence and associativity.
- Convert mathematical expressions from Infix to Postfix.
- Understand a practical application of stacks in processing heatwave monitoring calculations.